

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: pSV2neo, <i>Apa</i> L1 digest.
Species Used	Mouse, LM(TK-), connective tissue [L-M (TK-)].		

Before the Pulse

Cell Growth Medium	Dulbecco's MEM (GIBCO/BRL, Sigma)	Growth Phase at Harvest	50 to 75% confluent
		Pre-pulse Incubation	None

Wash Solution Phosphate Buffered Saline, without Ca++ or Mg++

The Pulse **Instruments Used** Gene Pulser® apparatus

Electroporation Temperature	4 °C		
Electroporation Medium*	Phosphate Buffered Saline, without Ca++ or Mg++	Cuvette Gap	0.4 cm
Cell Density	1.25 x 10 ⁷ cells / ml	Voltage	0.950 kV
Volume of Cells	0.8 ml	Field Strength	2.37 kV/cm
DNA Concentration	0.5 µg / ml	Capacitor	25 µF
DNA Resuspension Buffer	Phosphate Buffered sucrose	Resistor	(Pulse Controller) Ω none.
Volume of DNA	10 µl	Time Constant	0.4msec
After the Pulse			
Outgrowth Medium	Dulbecco's MEM		

Outgrowth Temperature	37 °C	Relevant Publications and/or Comments	
Length of Incubation	48 hours	Note: exponential values designated in parentheses.	
Selection Method or Assay Used	G-418 , 400 µg / ml after 48 hours recovery	**It is NOT RECOMMENDED to use high voltage with out the Pulse Controller.	
Electroporation Efficiency	1 x 10 ⁻³ transformants / cell		
Per Cent Survival	80 %		

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